

Randomized Control Trial

Jussi Keppo and Hong Ming Tan

National University of Singapore

DBA5101

What Is a Causal Effect?

A causal effect is defined as a comparison between two states of the world:

- (i) the actual state of the world;
- (ii) the alternative state of the world, the counterfactual state

What Is a Causal Effect? Cont'd

Think about a simple example:

- In the actual state of the world, a person takes aspirin for headache and one hour later reports the severity of the headache.
- In the counterfactual state of the world, that same person takes nothing for headache and one hour later reports the severity of the headache.

What Is a Causal Effect? Cont'd

Measure the causal effect of the aspirin:

- The causal effect of the aspirin is the difference in the severity of headache between the two states of the world.
- However, counterfactuals are not realized because they are hypothetical states of the world.
- **We have missing data on the counterfactual outcome!**

Treatment Variable

- A binary variable

$$D_i = \begin{cases} 1 & \text{if a particular unit } i \text{ receives the treatment} \\ 0 & \text{if it does not} \end{cases}$$

- Each unit will have two **potential outcomes**

$$\begin{cases} Y_i^1 & \text{if unit } i \text{ receives the treatment} \\ Y_i^0 & \text{if it does not, or say, is in the control state} \end{cases}$$

- Each unit i has an observable or **actual outcome** Y_i , and we have the switching equation:

$$Y_i = D_i Y_i^1 + (1 - D_i) Y_i^0$$

Treatment Effects

The unit-specific treatment effect, or causal effect, is given by

$$\delta_i = Y_i^1 - Y_i^0$$

Three Important Estimators

- ATE: the average treatment effect for the whole population
- ATT: the average treatment effect for the treatment group
- ATU: the average treatment effect for the untreated group (or control group)

Average Treatment Effect (ATE)

$$ATE = E[\delta_i] = E[Y_i^1 - Y_i^0] = E[Y_i^1] - E[Y_i^0]$$

- ATE is the average treatment effect for the whole population.
- Since we cannot observe both the actual and counterfactual worlds simultaneously, ATE is inherently unknowable.

Average Treatment Effect for Treatment Group (ATT)

$$\begin{aligned} ATT &= E[\delta_i | D_i = 1] = E[Y_i^1 - Y_i^0 | D_i = 1] \\ &= E[Y_i^1 | D_i = 1] - E[Y_i^0 | D_i = 1] \end{aligned}$$

- $D_i = 1$ refers to the units that are assigned to the treatment group.
- ATT is the population mean treatment effect for the group of units that had been assigned the treatment.
- ATT will differ from the ATE if D_i correlates with $Y_i^1 - Y_i^0$.
- ATT is also unknown because $Y_i^0 | D_i = 1$ is not observed.

Average Treatment Effect for Untreated Group (ATU)

$$\begin{aligned} ATU &= E[\delta_i | D_i = 0] = E[Y_i^1 - Y_i^0 | D_i = 0] \\ &= E[Y_i^1 | D_i = 0] - E[Y_i^0 | D_i = 0] \end{aligned}$$

- $D_i = 0$ refers to the units that are assigned to the control group.
- ATU is the population mean treatment effect for those units sorted into the control group.
- Typically in an observational setting $ATT \neq ATU$.
- ATU is unknown because $Y_i^1 | D_i = 0$ is not observed.

Example

Let's assume there are ten patients who have cancer, and two medical procedures or treatments.

- A surgery intervention, $D_i = 1$, and a chemotherapy intervention, $D_i = 0$, where i refers to the i 'th patient.
- Each patient has the following two potential outcomes, which are defined as post-treatment life span in years:
 - ▶ Y^1 is the life span in a world where they received surgery.
 - ▶ Y^0 is the life span in a world where they received chemo.
 - ▶ Y is the actual life span.

Table 1: Potential Outcomes

Patients	Y^1	Y^0	δ
1	7	1	6
2	5	6	-1
3	5	1	4
4	7	8	-1
5	4	2	2
6	10	1	9
7	1	10	-9
8	5	6	-1
9	3	7	-4
10	9	8	1

Note: In the real world, we cannot observe Y_1 and Y_0 simultaneously.

Example, Cont'd

Let's assume there exists a perfect doctor who knows each person's potential outcomes and chooses the treatment that maximizes the person's post-treatment life span.

Once the doctor makes the medical procedure decision, the actual outcomes are observed according to the switching equation:

$$Y_i = D_i Y_i^1 + (1 - D_i) Y_i^0$$

Table 2: Observed (Actual) Outcomes

Patients	Y	D
1	7	1
2	6	0
3	5	1
4	8	0
5	4	1
6	10	1
7	10	0
8	6	0
9	7	0
10	9	1

Example, Cont'd

Now using Tables 1 and 2, we get the three treatment effects:

- $ATE = E[Y_i^1] - E[Y_i^0]$
- $ATT = E[Y_i^1 | D_i = 1] - E[Y_i^0 | D_i = 1]$
- $ATU = E[Y_i^1 | D_i = 0] - E[Y_i^0 | D_i = 0]$

Example, Cont'd

Now using Tables 1 and 2, we get the three treatment effects:

- $ATE = E[Y_i^1] - E[Y_i^0]$
- $ATT = E[Y_i^1 | D_i = 1] - E[Y_i^0 | D_i = 1]$
- $ATU = E[Y_i^1 | D_i = 0] - E[Y_i^0 | D_i = 0]$

From the tables we get:

- $ATE = 0.6$
- $ATT = 4.4$
- $ATU = -3.2$

Note: In the real world, we cannot directly estimate these three treatment effects, because we do not know the counterfactual outcomes.

Example, Cont'd

Let's try to understand what these treatment effects tell us:

- $ATE = 0.6$ shows that, compared to the chemo, the average treatment effect of surgery across the patients is 0.6 additional years.
- $ATT = 4.4$ means that the average post-surgery life span for the surgery group is 4.4 additional years.
- $ATU = -3.2$ means that the average post-surgery life span for the control group is 3.2 fewer years.
 - ▶ Or in other words, they would have received 3.2 years shorter life if they had received surgery instead of chemo.

Note: The average treatment effect is 0.6, and there exist heterogeneous treatment effects.

Example, Cont'd

Remember that we have a missing data problem because the counterfactual outcome is unobservable!

One simple but commonly used estimator is called **the Simple Difference in Outcomes (SDO)**:

- It is defined as $E[Y^1|D = 1] - E[Y^0|D = 0]$
- From observational data:

$$\begin{aligned} SDO &= E[Y^1|D = 1] - E[Y^0|D = 0] \\ &= \frac{1}{N_T} \sum_{i=1}^{N_T} (y_i | D_i = 1) - \frac{1}{N_C} \sum_{i=1}^{N_C} (y_i | D_i = 0), \end{aligned}$$

where N_T and N_C are the numbers of units in the treatment and control groups.

Example, Cont'd

- By Table 2, the simple difference in outcomes $SDO = -0.4$.
- This indicates that surgery is harmful because the treatment group lives **0.4 fewer years** post-surgery than the chemo group.
- However, we know this is misleading.

Why does SDO give an incorrect treatment effect?

Where Does the Bias Come From?

- Patients were not randomly assigned to different treatment options.
- The doctor optimally sorted them into their best treatment options:
 - ▶ A patient received surgery if $Y^1 > Y^0$;
 - ▶ A patient received chemo if $Y^0 > Y^1$;
 - ▶ That means D depends on Y^1 and Y^0 .
- The treatment and control groups became fundamentally different, and these differences depended on the potential outcomes.

Where Does the Bias Come From? Cont'd

To make it more concrete, let's decompose SDO into three parts:¹

$$\begin{aligned} SDO &= E[Y^1|D=1] - E[Y^0|D=0] \\ &= ATE + E[Y^0|D=1] - E[Y^0|D=0] + (1 - \pi)(ATT - ATU), \end{aligned}$$

where π is the share of patients who received surgery and $1 - \pi$ is the share of patients who received chemotherapy.

¹The proof can be found in the appendix

Where Does the Bias Come From? Cont'd

To be more precise, we have:

$$\underbrace{\frac{1}{N_T} \sum_{i=1}^n (y_i \mid d_i = 1) - \frac{1}{N_C} \sum_{i=1}^n (y_i \mid d_i = 0)}_{\text{Simple Difference in Outcomes}} = \underbrace{E[Y^1] - E[Y^0]}_{\text{Average Treatment Effect}} + \underbrace{E[Y^0 \mid D = 1] - E[Y^0 \mid D = 0]}_{\text{Selection bias}} + \underbrace{(1 - \pi)(ATT - ATU)}_{\text{Heterogeneous treatment effect bias}}$$

Thus, the bias comes from **the selection bias and heterogeneous treatment effect bias**.

Understanding the Biases

So, we have:

$SDO = ATE + \text{Selection Bias} + \text{Heterogeneous Treatment Effect Bias}$,

where

- 1 Average Treatment Effect (ATE): the parameter of interest.
- 2 Selection bias: a description of the differences between the two groups **if there had never been a treatment**.
- 3 The third term is the bias of different treatment effects multiplied by the share of the chemo population.

Understanding the Biases, Cont'd

Let's verify the decomposition using the numerical example:

- ATE: 0.6
- The selection bias: -4.8
- Heterogeneous Treatment Effect Bias $(1 - \pi)(ATT - ATU)$:
 $0.5 \times [4.4 - (-3.2)] = 3.8$

$$\underbrace{\frac{1}{N_T} \sum_{i=1}^n (y_i | d_i = 1) - \frac{1}{N_C} \sum_{i=1}^n (y_i | d_i = 0)}_{\text{Simple Difference in Outcomes}} = \underbrace{E[Y^1] - E[Y^0]}_{\text{Average Treatment Effect}} + \underbrace{E[Y^0 | D = 1] - E[Y^0 | D = 0]}_{\text{Selection bias}} + \underbrace{(1 - \pi)(ATT - ATU)}_{\text{Heterogeneous treatment effect bias}}$$

Easy to verify that

$$-0.4 = 0.6 - 4.8 + 3.8$$

Can We Eliminate the Biases?

- Unfortunately, the two biases cannot be calculated as they depend on unobservable counterfactuals.
- However, we introduce an **independent assumption** to deal with the biases.

Independent Assumption

In our example, there was a perfect doctor who assigned patients to different groups based on their potential outcomes:

- A patient received surgery if $Y^1 > Y^0$;
- A patient received chemo if $Y^0 > Y^1$;
- That means D depends on Y^1 and Y^0 .

Independent Assumption, Cont'd

What if the treatment (surgery) was assigned to patients independent of their potential outcomes?

- For example, maybe the doctor alphabetized the patients by their last names, and the first five received surgery and the last five received chemotherapy.
- Notationally, this means $(Y^1, Y^0) \perp D$

Independent Assumption, Cont'd

If the doctor used a method for assigning treatment that is independent of the potential outcomes then:

- $E[Y^1|D = 1] - E[Y^1|D = 0] = 0$
- $E[Y^0|D = 1] - E[Y^0|D = 0] = 0$

That is, then each expected potential outcome (Y^1 or Y^0) is the same (in the population) for the surgery group and the chemotherapy group.

Independent Assumption, Cont'd

After such randomization, the selection bias and the heterogeneous treatment effect bias are both eliminated!

- Selection bias: $E[Y^0|D = 1] - E[Y^0|D = 0] = 0$
- $ATT = E[Y^1|D = 1] - E[Y^0|D = 1]$
- $ATU = E[Y^1|D = 0] - E[Y^0|D = 0]$
- Thus, the heterogeneous treatment effect $(1 - \pi)(ATT - ATU) = 0$.

If the treatment is independent of the potential outcomes
then $SDO = ATE!$

Independent Assumption, Cont'd

So, we need:

- Data on observable outcomes;
- Data on treatment assignment;
- The independent assumption: $(Y^1, Y^0) \perp D$.

SUTVA

Besides the independent assumption, we also need so-called **stable unit treatment value assumption (SUTVA)**:

- Each treated unit receives the same sized dose;
- **No spillovers ("externalities") to other units' potential outcomes;**
- No general equilibrium effects.

Randomized Control Trial Example

Team incentive and performance: Evidence from a retail chain²

- How can members of a team be rewarded and induced to work efficiently?
- Feibel et al. (2017) tried to answer this question by conducting a field experiment with a retail chain.
- The study examines the retail chain with 193 shops and 1,300 employees.

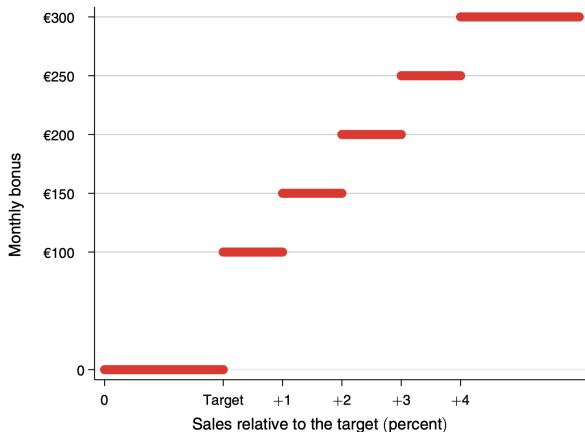
²<https://doi.org/10.1257/aer.20160788>

Design of the Field Experiment

- **Time and location:** The experiment was conducted from April to June 2014, across various locations in Germany where the retail chain operated.
- **Participants** were the sales teams working in the retail chain's shops. Each team had an average of 7 employees.
- The researchers **randomly assigned** these shops into two groups: the treatment group and the control group.
 - ▶ **Treatment group** consisted of 97 shops where employees were eligible to receive a team-based bonus if their shop's sales exceeded pre-determined monthly targets.
 - ▶ **Control group** consisted of 96 shops that continued their normal operations without any additional incentives.

Design of the Field Experiment, Cont'd

The bonus scheme:



- The bonus was distributed among the team members (excluding mini-jobbers) in proportion to their working hours during the month.

Design of the Field Experiment, Cont'd

Information and training:

- Before the experiment, the district managers were informed and trained on the bonus scheme.
- They then communicated this information to shop supervisors in the treatment group.
- Employees were not made aware that they were part of an experiment; instead, the incentive scheme was presented as a regular company initiative.

Does the Independent Assumption Hold?

- **Randomization:** Shops were randomly assigned to the treatment (team bonus) or control group, ensuring that any differences in performance could be attributed to the incentive rather than other factors.
 - ▶ Feiebel et al. (2017) ranked shops based on predicted sales and then paired and randomly assigned them to treatment or control groups.
 - ▶ The randomization was verified by checking the balance in pre-treatment characteristics between the two groups.
- **How about SUTVA?**

Does the Independent Assumption Hold? Cont'd

TABLE 1—PRETREATMENT SHOP CHARACTERISTICS

	All shops (<i>N</i> = 193)	Control (<i>N</i> = 96)	Treatment (<i>N</i> = 97)	Diff: control versus treatment	<i>t</i> -test <i>p</i> -value
<i>Panel A. Quantitative performance indicators</i>					
Mean monthly sales	27,820 (13,094)	27,453 (11,481)	28,194 ^a (14,542)	741 (1,890)	0.695
Mean monthly sales (in logs)	10.15 (0.40)	10.14 (0.39)	10.15 (0.41)	0.01 (0.06)	0.846
Sales trends (year-on-years sales growth)	−0.04 (0.13)	−0.04 (0.13)	−0.03 (0.12)	0.01 (0.01)	0.322
Unsold goods as percent of sales	16.04 (6.80)	16.16 (7.01)	15.92 (6.90)	−0.24 (0.60)	0.694
Mean number of customer visits	10,079 (3,969)	10,028 (3,921)	10,131 (4,018)	103 (566)	0.856
Frequency of achieving the sales target	35.5%	35.8%	35.2%	−0.6%	0.860
<i>Panel B. Qualitative performance indicators</i>					
Mean mystery shopping score	97.6%	97.6%	97.6%	0.0%	0.826
<i>Panel C. Locations of shops</i>					
Big town	30.0%	33.3%	26.8%	−6.5%	0.194
Mean monthly property rent per m ² (zip code)	8.79 (1.84)	8.94 (1.85)	8.72 (1.87)	−0.22 (0.27)	0.432
Total number of other bakeries (1 km radius)	2.67 (2.75)	2.81 (2.88)	2.52 (2.60)	−0.30 (0.40)	0.462
Total number of Aldi/Lidl shops (1 km radius)	0.89 (0.90)	0.89 (0.92)	0.90 (0.87)	0.01 (0.13)	0.932

Does the Independent Assumption Hold? Cont'd

TABLE 2—PRETREATMENT HR CHARACTERISTICS

	All shops (N = 193)	Control (N = 96)	Treatment (N = 97)	Diff: control versus treatment	t-test p-value
<i>Panel A. Characteristics of shop supervisors</i>					
Mean weekly working hours	34.3 (4.3)	34.5 (4.0)	34.1 (4.5)	−0.4 (0.6)	0.532
Mean age, years	41.4	40.9	41.9	0.9	0.534
Share of females	94.7%	98.6%	90.7%	−7.9%	0.012
Mean monthly quit rate	0.7%	0.6%	0.7%	0.1%	0.691
<i>Panel B. Characteristics of regular sales agents (excluding mini-jobbers)</i>					
Mean number per shop	3.7 (2.0)	3.7 (2.0)	3.7 (1.9)	−0.1 (0.3)	0.770
Mean weekly working hours	26.5 (5.1)	26.7 (5.1)	26.2 (5.1)	−0.5 (0.4)	0.217
Mean age, years	40.5	40.3	40.6	0.3	0.768
Share of females	93.1%	92.5%	93.7%	1.3%	0.410
Mean monthly quit rate	2.5%	2.6%	2.4%	0.2%	0.458
Share of full-time agents (>35 hrs/week)	14.8%	14.7%	14.8%	0.1%	0.965
Share of agents with permanent contract	65.2%	64.0%	66.3%	2.3%	0.510
Share of agents without professional training in retail	63.5%	66.3%	60.8%	−5.5%	0.131

Regression Model

Because the data is generated through a random assignment, the following regression gives causality:

$$\ln(sales_{it}) = \beta \cdot treatment_i + month\ fixed\ effect_t + controls_{it} + error_{it},$$

where

- i is the shop index, and t refers to time. The data is collected monthly.
- **Control variables:** The model controlled for time-variant variables such as total hours worked, shop refurbishment, and month-fixed effects.
- **Main results:** The team bonus **increased sales** in the treated shops by **around 3 percent**, with no significant increase in sales per customer, indicating that the increase was due to higher customer throughput rather than individual customer spending.

Results

Table: Average Treatment Effect, April–June 2014

	$\ln(\text{sales}_{it})$
Treatment_i	0.032 (0.014)
Month fixed effects	Yes
Controls	Yes
Shop fixed effects	No
Average $\ln(\text{sales}_{it})$ pretreatment	Yes
Observations	576

Managerial Implications

- **Profitability:** The bonus scheme was highly profitable, generating €3.80 in sales and €2.10 in profit for every euro spent on bonuses.
- **Long-term viability:** The positive effects were stable throughout the experiment, and the company decided to roll out the bonus scheme to all shops, leading to a long-term increase in performance.
- **Team dynamics:** The experiment demonstrated that team-based incentives could effectively increase performance, especially in settings where teamwork and mutual support are essential.

Appendix: Simple Difference in Outcomes (SDO)

Simple difference in expectations:

$$\begin{aligned} & E[Y^1 \mid D = 1] - E[Y^0 \mid D = 0] \\ &= ATE + E[Y^0 \mid D = 1] - E[Y^0 \mid D = 0] + (1 - \pi)(ATT - ATU) \end{aligned}$$

Appendix: SDO, Cont'd

To understand where these parts on the right-hand side originate, we need to decompose the parameter of interest, ATE, into its basic building blocks. ATE is equal to the weighted sum of conditional average expectations, ATT and ATU:

$$\begin{aligned}ATE &= \pi ATT + (1 - \pi) ATU \\&= \pi E[Y^1 \mid D = 1] - \pi E[Y^0 \mid D = 1] \\&\quad + (1 - \pi) E[Y^1 \mid D = 0] - (1 - \pi) E[Y^0 \mid D = 0] \\&= \left\{ \pi E[Y^1 \mid D = 1] + (1 - \pi) E[Y^1 \mid D = 0] \right\} \\&\quad - \left\{ \pi E[Y^0 \mid D = 1] + (1 - \pi) E[Y^0 \mid D = 0] \right\}\end{aligned}$$

where π is the share of patients who received surgery and $1 - \pi$ is the share of patients who received chemotherapy.

Appendix: SDO, Cont'd

Because the conditional expectation notation is a little cumbersome, let's introduce the following:

$$E[Y^1 \mid D = 1] = a$$

$$E[Y^1 \mid D = 0] = b$$

$$E[Y^0 \mid D = 1] = c$$

$$E[Y^0 \mid D = 0] = d$$

$$ATE = e$$

Appendix: SDO, Cont'd

Now we can write the ATE as follows:

$$e = \{\pi a + (1 - \pi)b\} - \{\pi c + (1 - \pi)d\}$$

$$e = \pi a + b - \pi b - \pi c - d + \pi d$$

$$e = \pi a + b - \pi b - \pi c - d + \pi d + (a - a) + (c - c) + (d - d)$$

$$0 = e - \pi a - b + \pi b + \pi c + d - \pi d - a + a - c + c - d + d$$

$$a - d = e - \pi a - b + \pi b + \pi c + d - \pi d$$

$$a - d = e + (c - d) + a - \pi a - b + \pi b - c + \pi c - d$$

$$a - d = e + (c - d) + (1 - \pi)a - (1 - \pi)b + (1 - \pi)d - (1 - \pi)c$$

Appendix: SDO, Cont'd

Thus, we have:

$$\begin{aligned} & E[Y^1 \mid D = 1] - E[Y^0 \mid D = 0] \\ &= ATE + \left(E[Y^0 \mid D = 1] - E[Y^0 \mid D = 0] \right) + (1 - \pi)(ATT - ATU), \end{aligned}$$

which is the SDO formula.